

[Aim: 100/100 in Maths]

आरंभ

TRIGONOMETRY

Lecture #5

ATK²:

Reciprocal
identities

$$\left\{ \begin{array}{l} \sin \theta \longleftrightarrow \frac{1}{\operatorname{cosec} \theta} \\ \cos \theta \longleftrightarrow \frac{1}{\sec \theta} \\ \tan \theta \longleftrightarrow \frac{1}{\cot \theta} \end{array} \right.$$

$$\left\{ \begin{array}{l} \tan \theta = \frac{\sin \theta}{\cos \theta} \\ \cot \theta = \frac{\cos \theta}{\sin \theta} \end{array} \right.$$

~~$\tan A = \cot B$~~

~~$A = B$~~

$\tan A = \tan B$
on comparing

$\angle A = \angle B$

#Trigonometric Identities:

[Square ~~diff~~ identities]

✓ $\sin^2 \theta + \cos^2 \theta = 1$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\underline{\underline{\sec^2 \theta = 1 + \tan^2 \theta}}$$

✓ $\sec^2 \theta - \tan^2 \theta = 1$

$$\sec^2 \theta - 1 = \underline{\underline{\tan^2 \theta}}$$

✓ $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

$$\underline{\underline{\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta}}$$

$$\operatorname{cosec}^2 \theta - 1 = \underline{\underline{\cot^2 \theta}}$$

Prove that

#LP: $(1 + \cot^2 A) \sin^2 A = 1$ ✓

LHS

LHS $\Rightarrow (1 + \cot^2 A) \sin^2 A$

$\Rightarrow (\operatorname{cosec}^2 A) \sin^2 A$

$\Rightarrow \frac{1}{\sin^2 A} \times \cancel{\sin^2 A}$

$\Rightarrow 1 = \text{LHS}$

~~HP~~

Sq. aTmī identities

$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

$\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$

Proof aTmī Q??

①

LHS



= RHS

②

RHS



LHS =

③

LHS



RHS



=

#LP: $\operatorname{cosec} \theta \sqrt{1 - \cos^2 \theta} = 1$

कट कर दिया जाएगा

(LHS) $\rightarrow$

$$\frac{\operatorname{cosec} \theta}{\sin \theta} \sqrt{1 - \cos^2 \theta}$$

$$\Rightarrow \frac{1}{\sin \theta} \times \sqrt{\sin^2 \theta}$$

$$\Rightarrow \frac{1}{\sin \theta} \times \sin \theta$$

$$\Rightarrow 1 = \text{RHS} \quad (\text{H.P.})$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

#LP: $(\sec^2 \theta - 1)(\operatorname{cosec}^2 \theta - 1) = 1$

LHS $\rightarrow (\sec^2 \theta - 1)(\operatorname{cosec}^2 \theta - 1)$

$\downarrow$ $\downarrow$

$[\tan^2 \theta]$ $[\cot^2 \theta]$

$\Rightarrow \frac{1}{\cancel{\cot^2 \theta}} \times \cancel{\cot^2 \theta}$

$\Rightarrow 1 = \text{RHS}$ ~~HP~~

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\boxed{\sec^2 \theta - 1 = \tan^2 \theta}$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\boxed{\operatorname{cosec}^2 \theta - 1 = \cot^2 \theta}$$

#LP: $\tan \theta + \frac{1}{\tan \theta} = \sec \theta \operatorname{cosec} \theta$

let $\sin \theta = s$
 $\cos \theta = c$

$\frac{1}{\cos} \rightarrow \frac{1}{\sin}$

RHS में sin, cos types दिए हैं $\frac{1}{s}$
 LHS में भी sin, cos बता दी

(LHS) $\frac{s}{c} + \frac{1}{\frac{s}{c}}$

$\Rightarrow \frac{s}{c} + \frac{c}{s}$

$\rightarrow \frac{s^2 + c^2}{c \cdot s}$

$\Rightarrow \frac{1}{c \cdot s} \left(\because \sin^2 \theta + \cos^2 \theta = 1 \right)$

$\Rightarrow \frac{1}{\cos \theta} \cdot \frac{1}{\sin \theta}$

$\Rightarrow \sec \theta \operatorname{cosec} \theta$ HP

#LP:

$$\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}$$

let $\cos A = c$, $\sin A = s$

LHS

sin, cos of
convert

$$\begin{aligned} & \Rightarrow \frac{\frac{c}{s} - c}{\frac{c}{s} + c} \\ & \Rightarrow \frac{c - c \cdot s}{c + c \cdot s} \\ & \Rightarrow \frac{c(1 - s)}{c(1 + s)} \end{aligned}$$

RHS में sab sin में है
(CBSE 2005)

$$\Rightarrow \frac{1 - \sin A}{1 + \sin A}$$

$$\Rightarrow \frac{1 - \frac{1}{\operatorname{cosec} A}}{1 + \frac{1}{\operatorname{cosec} A}}$$

$$\Rightarrow \frac{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A}}{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A}}$$

$$\Rightarrow \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1} = \text{RHS}$$

HP

#LP: $(\operatorname{cosec} A + \cot A)(1 - \cos A) = \sin A$

LHS $\rightarrow \left(\frac{1}{s} + \frac{c}{s} \right) (1 - c)$

$\rightarrow \left(\frac{1+c}{s} \right) (1-c)$

$\rightarrow \frac{(1)^2 - (c)^2}{s}$

$\rightarrow \frac{1 - \cos^2 \theta}{\sin \theta} \Rightarrow \frac{\sin^2 \theta}{\sin \theta} \Rightarrow \sin \theta = \text{RHS}$ HP

RHS $\rightarrow \sin$

LHS को $\sin / \cos$ में बदलें

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

#LP: $\sec A (1 - \sin A) (\sec A + \tan A) = 1$

LHS

$$(\sec A - \sec A \cdot \sin A) (\sec A + \tan A)$$

$$\left(\sec A - \frac{1}{\cos A} \sin A \right) (\sec A + \tan A)$$

$$(\sec A - \tan A) (\sec A + \tan A)$$

$$\sec^2 A - \tan^2 A$$

$$\Rightarrow 1 = \text{RHS}$$

#LP: $\underline{(1 + \tan^2 \theta)} \underline{(1 - \sin \theta)} \underline{(1 + \sin \theta)} = 1$

LHS $(\sec^2 \theta) (1^2 - \sin^2 \theta)$

$(\sec^2 \theta) (\cos^2 \theta)$

$\frac{1}{\cancel{\cos^2 \theta}} \times \cancel{\cos^2 \theta}$

$\Rightarrow \textcircled{1} = \text{RHS} \quad \textcircled{\text{HP}}$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\boxed{\sec^2 \theta = 1 + \tan^2 \theta}$$

#LP: $(\sec \theta + \cos \theta)(\sec \theta - \cos \theta) = \tan^2 \theta + \sin^2 \theta$

$(a+b) \quad (a-b)$

LHS

$\sec^2 \theta \leftrightarrow \tan^2 \theta$

$\sec^2 \theta - \cos^2 \theta$

$\rightarrow (1 + \tan^2 \theta) - (1 - \sin^2 \theta)$

$\cos^2 \theta \leftrightarrow \sin^2 \theta$

$\rightarrow \cancel{1} + \tan^2 \theta - \cancel{1} + \sin^2 \theta$

$\rightarrow \boxed{\tan^2 \theta + \sin^2 \theta} = \text{RHS}$

HP

$\sec^2 \theta - \tan^2 \theta = 1$

$\boxed{\sec^2 \theta = 1 + \tan^2 \theta}$

$\sin^2 \theta + \cos^2 \theta = 1$

$\boxed{\cos^2 \theta = 1 - \sin^2 \theta}$

THANK YOU COODIESS !!

